Chapter 13 • Probability

Solved Exercise 13.1

1. A die is labelled L,M,N,O,P,U. Find P(consonant).

Consonants L,M,N,P are four of six equally likely letters: $P=\frac{4}{6}=\frac{2}{3}$

2. Two fair dice are thrown. Find the four probabilities.

(i) Sum at least 4: exclude sums 2 and 3, which have 1+2=3 outcomes. $P=\frac{36-3}{36}=\frac{11}{12}$

(ii) Product from 5 to 10 inclusive: product 5 gives 2 outcomes, 6 gives 4, 7 gives 0, 8 gives 2, 9 gives 1, 10 gives 2; total 11. $P=\frac{11}{36}$

(iii) Difference 4: (1,5),(5,1),(2,6),(6,2), so $P = \frac{4}{36}$ $= \frac{1}{9}$

(iv) 6 outcomes. $P = \frac{6}{36}$ $= \frac{1}{6}$

3. One letter is selected at random from MATHEMATICS. Find the six requested probabilities.

The eleven letters are M,A,T,H,E,M,A,T,I,C,S. Thus there are four vowels, seven consonants, one E, two A's, two M's and two T's.

(i) $P(\text{vowel})=\frac{4}{11}$

(ii) $P(\text{consonant})=\frac{7}{11}$

(iii) $P(E)=\frac{1}{11}$

(iv) $P(A)=\frac{2}{11}$

(v) $P(\text{not M})=1-\frac{2}{11}=\frac{9}{11}$

(vi) $P(\text{not T})=1-\frac{2}{11}=\frac{9}{11}$

4. A fair die is rolled. Find P(3 or 4) and its complement.

$P(3 \text{ or} 4)=\frac{2}{6}=\frac{1}{3}$; $P(\text{not } 3 \text{ or} 4)=1-\frac{1}{3}=\frac{2}{3}$

5. Cards numbered 1–30 are selected at random.

(i) Card 25: $\frac{1}{30}$

(ii) Inclusive 17–22: six numbers, so $\frac{6}{30}=\frac{1}{5}$

(iii) At least 20: 20–30 gives 11 numbers, so $\frac{11}{30}$

(iv) Not 27 and not 29: exclude two cards, so $\frac{28}{30}=\frac{14}{15}$

(v) Not in inclusive 12–15: exclude four cards, so $\frac{26}{30}=\frac{13}{15}$

6. P(Ayesha passes)=0.85. Find P(she does not pass).

1−0.85=0.15.

7. A fair coin and fair die are used once; the joint sample space has 12 outcomes.

(i) Tail with 4,5 or6: $\frac{3}{12}=\frac{1}{4}$

(ii) Head with 2 or3: $\frac{2}{12}=\frac{1}{6}$

(iii) Literal "head and tail" on one coin with 6 is impossible: 0. If "head or tail" was intended, any coin face with 6 gives $\frac{1}{6}$, matching the key.

studysapiens.com

(iv) Key's intended complement of (tail and5): $1-\frac{1}{12}=\frac{11}{12}$ If only "not tail" is negated, head and5 gives $\frac{1}{12}$

(v) Key's intended complement of (head and 5-or-2): $1-\frac{2}{12}=\frac{5}{6}$ If only "not head" is negated, tail and 5-or-2 gives $\frac{2}{12}=\frac{1}{6}$

8. From a 52-card pack, find P(queen) and P(neither queen nor jack).

(i) $\frac{4}{52}=\frac{1}{13}$

(ii) exclude four queens and four jacks: $\frac{44}{52}=\frac{11}{13}$

9. From a 52-card pack, find P(jack) and P(no diamond).

(i) $\frac{4}{52}=\frac{1}{13}$

(ii) non-diamonds = $52-13$

$= 39$, so $\frac{39}{52}$

$= \frac{3}{4}$

studysapiens.com